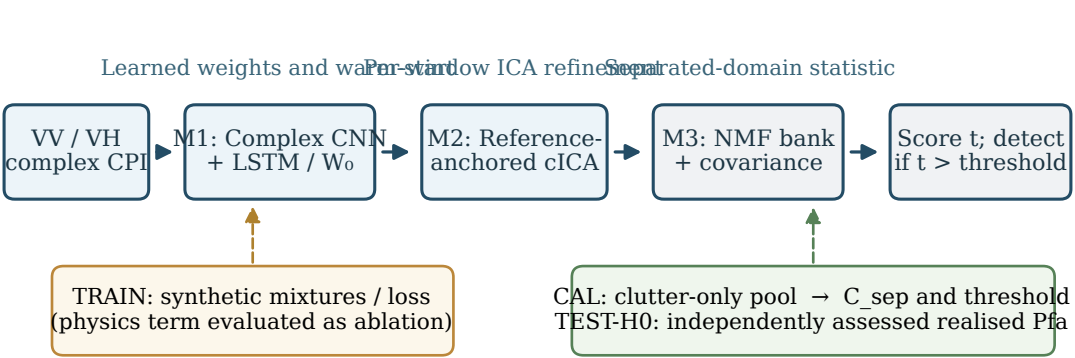
Network initialization is not a detector decision
Learned weights and warm-start
Per-window ICA refinement
Separated-domain statistic
VV / VH complex CPI
M1: Complex CNN + LSTM / W0
M2: Reference-anchored cICA
M3: NMF bank + covariance
Score t; detect if t > threshold
TRAIN: synthetic mixtures / loss (physics term evaluated as ablation)
CAL: clutter-only pool → C_sep and threshold
TEST-H0: independently assessed realised Pfa